PhyST-Net: An Exogenous-Aware Spatio-Temporal Graph Transformer for Domain-Constrained Network Forecasting

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***Abstract*—Spatio-temporal forecasting is a core task for graph-structured networks such as power grids, traffic systems, and industrial sensor clusters, where accurate node-level prediction requires joint modeling of historical trends, external drivers, and operational constraints. Existing forecasting models often treat historical covariates, future exogenous variables, and domain priors as isolated inputs, limiting their ability to capture the interplay between temporal continuity, dynamic spatial topology, and physical feasibility. This paper proposes PhyST-Net, an exogenous-aware spatio-temporal graph Transformer designed as a general framework for domain-constrained network forecasting. PhyST-Net formulates prediction as a multi-modal graph learning task with historical target states, historical exogenous variables (ExG), future ExG, and graph metadata as distinct interfaces. The framework employs: (1) Adaptive Gaussian-soft-windowed Patching (AGSP) for continuous temporal tokenization; (2) a Relational Spatio-Temporal Manifold Fusion (R-STMF) graph adapter for multi-scale spatial dependency modeling; (3) KAN-enhanced Adaptive Conditioning (K-AC) for non-linear exogenous modulation; and (4) a Domain-Informed Dynamic Capping Head (DI-DCH) for constraint-aware decoding. While the framework is general, we validate its effectiveness on the challenging task of wind power forecasting using the SD-WPF, HoT, and Kelmarsh datasets. Results show that PhyST-Net consistently improves forecasting accuracy and feasibility across multiple horizons, demonstrating the value of explicit exogenous awareness and domain-informed architectural inductive biases.**

***Index Terms*—Spatio-temporal forecasting, graph Transformers, exogenous variables, domain-constrained machine learning, network forecasting.**

I. Introduction

SPATIO-TEMPORAL forecasting predicts future states in graph-structured systems (e.g., transportation networks, power grids) where temporal dynamics and spatial interactions evolve jointly. Node-level trajectories are rarely self-driven; rather, they are shaped by historical conditions, future external drivers, and domain-specific feasibility constraints. As illustrated in Fig. 1, complex network signals comprise shared macro-trends driven by exogenous forcing and high-frequency local variations. Consequently, a practical forecasting model must learn continuous temporal representations, infer dynamic graph relations, and integrate future exogenous variables (ExG) without data leakage, all while maintaining operational compatibility.

A. The Forecasting Trilemma in Spatio-Temporal Networks

Despite recent architectural advancements, modern spatio-temporal forecasting frameworks remain hampered by a "Forecasting Trilemma"—an inherent trade-off between Model Capacity, Domain Consistency, and Spatio-Temporal Complexity.

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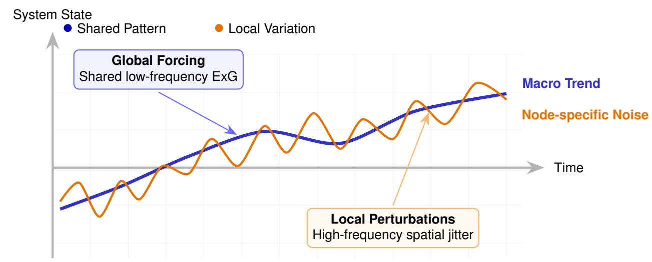

Fig. 1. Multi-scale spatio-temporal processes. General network signals are composed of shared macro-trends driven by exogenous forcing and high-frequency variations unique to individual nodes. PhyST-Net separates these components through dual-branch graph reasoning.

1) Model Capacity vs. High-Frequency Noise: Capacity denotes architectural expressive power. While modern Transformers excel at sequence modeling, they are highly sensitive to high-frequency stochastic noise in raw spatio-temporal data. Furthermore, rigid temporal patching exacerbates this by artificially splitting continuous physical regimes. As highlighted by recent trends in complexity-balanced sequence modeling, uniform representational capacity across an entire time horizon is computationally inefficient. The challenge is designing an architecture capable of capturing macro-trends while employing adaptive temporal allocation to filter node-specific noise.

2) Domain Consistency and Constraint-Aware Operations: Consistency mandates that a model's outputs adhere to the fundamental physical laws or operational limits inherent to the target domain. Purely data-driven architectures frequently generate unfeasible predictions—such as negative vehicular speeds or power generation exceeding a facility's nameplate capacity. Such violations severely degrade operator trust in critical infrastructure. Inspired by emerging paradigms in safety-critical generation that utilize output-space guidance, a robust forecasting framework must enforce physical feasibility directly within its decoding pathway, moving beyond the limitations of reactive, post-hoc corrections.

3) Spatio-Temporal Complexity and Multi-Scale Dynamics: Complexity involves the management of dense spatial couplings across heterogeneous nodes. In physical networks, nodes interact via multiple simultaneous mechanisms: local topological dependencies (e.g., aerodynamic wake effects, adjacent road segments) and global environmental fronts (e.g., synoptic weather patterns). Existing models frequently conflate these multi-scale interactions into a singular spatial operator, thereby failing to distinguish localized perturbations from system-wide drivers.

To systematically resolve this trilemma, we propose PhyST-Net, an ExG-aware Spatio-Temporal Graph Transformer designed as a versatile framework for domain-constrained network forecasting. To accommodate the multi-scale and multi-modal nature of dynamic network signals, we introduce the following key innovations:

- **Continuous and Differentiable Patching**: We present the **Adaptive Gaussian-soft-windowed Patching (AGSP)** mechanism. By transforming discrete observations into learnable spectral-temporal densities, AGSP recovers continuous event dynamics and functions as an adaptive low-pass filter, effectively rejecting high-frequency noise.
- **Decoupled Spatio-Temporal Topology**: We introduce the **Relational Spatio-Temporal Manifold Fusion (R-STMF)** framework. R-STMF leverages a dual-branch graph logic to decompose latent node dynamics into multi-scale spatial manifolds, mitigating the spatial conflation typical of standard distance-based graphs.
- **Piecewise Conditioning and Safe Capping**: We implement the **KAN-enhanced Adaptive Meteorological Conditioning (K-AMC)** module, utilizing Kolmogorov-Arnold spline edge functions for the non-linear modulation of graph-enhanced states via exogenous drivers. This is coupled with a **Physics-Informed Dynamic Capping Head (PI-DCH)** that embeds strict physical safety constraints and adherence to operational envelopes directly into the output space.

By explicitly partitioning the data flow into distinct computational pathways—historical context, relational reasoning, and exogenous conditioning—PhyST-Net establishes a rigorous forecasting contract. This interface-centric design prevents the entanglement of historical data and future drivers, ensuring the architecture can be seamlessly instantiated across diverse critical domains.

B. Contributions

The primary contributions of this work are summarized as follows:

- **A general ExG-aware graph-forecasting formulation.** We define network forecasting as a multi-modal task where historical states, exogenous variables, and domain constraints are processed through distinguishable pathways, thereby supporting broad architectural generalizability.
- **A continuous temporal tokenization module.** We introduce AGSP, which supersedes rigid patch boundaries with learnable Gaussian soft-windows, preserving event continuity and minimizing boundary artifacts in non-stationary sequences.
- **A modular graph adapter for mixed spatial mechanisms.** We develop R-STMF to disentangle multi-scale spatial dependencies prior to relational propagation, providing a modular contract that accommodates alternative graph predictors.
- **A future-ExG-conditioned decoding path.** We fuse K-AMC and PI-DCH to inject forecast-horizon drivers and structurally enforce feasibility constraints within the model, bridging latent representation learning with strict operational requirements.

The effectiveness and robustness of PhyST-Net are validated through comprehensive experiments on real-world datasets spanning diverse scales and operational conditions, confirming its viability as a state-of-the-art general-purpose network forecasting framework.

II. Related Work

This section reviews representative research trajectories pertinent to PhyST-Net. Rather than chronicling models chronologically, we categorize prior literature based on the core modeling challenges inherent to ExG-aware spatio-temporal forecasting: temporal dependency modeling, spatial graph learning, exogenous-variable fusion, and constraint-aware output generation.

A. Temporal Forecasting and Transformer Architectures

Classical statistical forecasting techniques, including autoregressive and state-space models, retain utility when predicting low-dimensional, approximately stationary processes. However, these foundational assumptions become restrictively fragile in graph-structured systems where node dynamics are highly non-linear and persistently coupled with external forcing. While deep recurrent and convolutional architectures improved non-linear sequence modeling via latent state tracking and local temporal filtering, long-horizon forecasting remains challenging due to error accumulation and vanishing gradients over extended time steps.

Transformer-based forecasters resolve long-range dependency constraints via self-attention, establishing themselves as the dominant paradigm for deep time-series analysis. Early breakthrough models, such as Informer [2], Autoformer [3], and FEDformer [4], focused on reducing the quadratic complexity of attention and decomposing trend-seasonal patterns. Recent variants, utilizing PatchTST-style patching and inverted-dimension attention, significantly reduce computational overhead while enhancing representation learning for multivariate sequences [1], [5], [6]. Nevertheless, temporal patching is predominantly implemented via rigid, fixed-length windows. In highly non-stationary sequences, physical regime transitions and rapid fluctuations rarely align with these artificial boundaries. PhyST-Net advances this trajectory by introducing AGSP, rendering temporal tokenization fully learnable and mathematically continuous.

B. Spatio-Temporal Graph Forecasting

Spatio-temporal graph neural networks (STGNNs) build upon foundational graph convolutions [7] and graph attention mechanisms [8] to capture node interactions. These methodologies are highly effective when cross-node dependencies dictate system behavior, such as in traffic corridors and power grids [9]–[11]. Furthermore, adaptive graph learning relaxes the restrictive assumption of a static, fully observable topology, allowing models to infer latent relations directly from data and dynamically adjust propagation patterns.

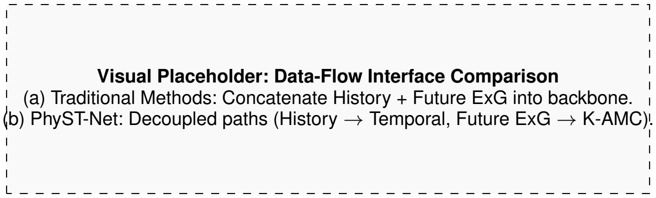

Fig. 2. Comparison of Exogenous Variable Fusion Interfaces. (a) Standard forecasting models frequently concatenate historical data and future drivers early in the network, increasing the risk of temporal leakage and confounding latent representation learning. (b) PhyST-Net explicitly decouples these data flows, reserving future drivers exclusively for horizon-level conditioning via the K-AMC module.

A critical limitation, however, is the widespread reliance on homogeneous propagation protocols across all latent states. In physical systems, spatial dependence is inherently multi-modal; nodes interact through localized proximities, shared environmental influences, and overarching exogenous drivers. A singular adjacency matrix, even if learned dynamically, conflates these distinct mechanisms. R-STMF resolves this by decomposing temporal representations into high-frequency local variations and low-frequency macro-trends, applying targeted, sparse relational propagation to each manifold prior to fusion.

C. Exogenous-Variable-Aware Forecasting

Real-world forecasting is intrinsically dependent on exogenous (ExG) variables. In vertical domains, such as intelligent traffic routing—where models like ConFormer [12] leverage external factors like accidents and regulations—and renewable energy dispatch, ExG features are critical for predictive resilience.

Recent foundational architectures have begun systematically targeting ExG fusion to address inconsistent covariate effects [13]. In time-series modeling, TimeXer [14] empowers Transformers by mapping external factors into variate tokens, enabling cross-attention between endogenous patches and exogenous drivers. Concurrently, DAG [15] introduces dual correlation networks to capture channel-wise dependencies, while in the spatio-temporal graph domain, models like GCGNet [16] employ generative frameworks to align predicted correlations with joint temporal-channel graphs driven by external factors.

While these mechanisms significantly improve accuracy under ideal conditions, they frequently blur the temporal distinction between historical observations and future drivers. Specifically, historical ExG elucidates recent state trajectories, whereas future ExG provides horizon-level modulation. Merging both into a singular feature tensor exacerbates temporal imbalance effects [13] and complicates data availability reasoning at inference time. Consequently, PhyST-Net structurally decouples historical ExG from future ExG (as illustrated in Fig. 2). K-AMC leverages the future-ExG pathway to generate latent modulation parameters, enabling forecast-horizon drivers to dynamically condition graph-enhanced states without being erroneously treated as historical priors.

D. Interpretable Conditional Modeling with KANs

Interpretability is paramount in operational forecasting, as dispatchers require insight into the primary drivers shaping a prediction. While attention weights and post-hoc attribution methods offer diagnostic utility, they often fail to reflect the model's actual computational pathways. Kolmogorov-Arnold Networks (KAN) [17] represent a paradigm shift from traditional perceptrons by modeling transformations via learnable, univariate spline-based edge functions [18]–[21].

By structuring a conditioning network as an additive sum of feature-wise edge functions, the influence of individual input variables can be analytically inspected through learned basis coefficients and derivatives. PhyST-Net capitalizes on this property within the K-AMC module: future ExG features are mapped to scale and shift parameters via spline functions. While not establishing strict causal identifiability, this architectural alignment provides a transparent, inspectable explanation mechanism far superior to generic dense concatenation.

E. Scalable Spatio-Temporal Forecasting

Computational scalability is a persistent bottleneck in graph forecasting. Dense spatio-temporal attention scales quadratically with both look-back length and graph cardinality. Conventional mitigation strategies—such as temporal down-sampling, rigid patching, and aggressive graph truncation—improve efficiency but risk discarding critical physical dynamics.

PhyST-Net approaches scalability dualistically. First, AGSP compresses the raw temporal dimension into a highly dense sequence of continuous tokens, preserving soft event boundaries while drastically lowering attention overhead. Second, R-STMF employs a sparse Top-K graph propagation scheme, ensuring that node interaction costs scale relative to a parameterized neighborhood size rather than the complete graph matrix. The theoretical complexity analysis detailed in Section IV-G connects these design choices directly to practical deployment constraints.

F. Physics-Informed and Constraint-Aware Forecasting

Physics-informed learning embeds essential domain knowledge into neural architectures via structural priors, penalty losses, or projection operators. In network forecasting, enforcing such constraints is vital, as system targets are strictly bounded by operational capacities. Unfeasible predictions, regardless of aggregate error metrics, can trigger catastrophic downstream decisions.

Contemporary constraint-aware frameworks predominantly rely on physics-guided loss regularization [22], [23] or reactive post-hoc clipping [24], [25]. However, soft penalties do not guarantee compliance, and post-hoc clipping creates a structural disconnect between the trained model and inference behavior. Aligning with the philosophy of structural physics induction [26], the PI-DCH layer internalizes feasibility operations directly within the forecasting head. By utilizing smooth barrier functions to shape predictions toward a dynamic capacity envelope prior to a final non-linear projection, the constraint pathway becomes an explicit, differentiable component of the computational graph.

G. Position of This Work

PhyST-Net synthesizes advancements in Transformer forecasting, adaptive graph learning, and interpretable exogenous conditioning. Its distinguishing factor lies in the rigorous organization of the forecasting interface. Rather than presenting a bespoke, single-domain architecture, this paper formulates a generalized ExG-aware graph prediction framework, subsequently instantiating physically meaningful, domain-specific modules. This methodology ensures PhyST-Net serves both as a state-of-the-art general-purpose model and as a robust, reusable template for domain-constrained network forecasting.

III. Problem Definition

We formulate the task as ExG-aware spatio-temporal graph forecasting. Let $G = (V, E, A)$ denote a graph, where $V = \{v_1, \ldots, v_N\}$ is the node set, E is the edge set, and $A \in \mathbb{R}^{N\times N}$ is an adjacency or relation matrix. A node can represent a network element, road sensor, grid bus, monitoring station, or industrial device depending on the application. The graph relation A may be physical, geographic, statistical, learned, or a combination of these sources.

At forecasting origin t, the historical target tensor is

$$\mathbf{X}_{t-T+1:t} \in \mathbb{R}^{T\times N\times D_x}, \tag{1}$$

where T is the look-back length, N is the number of nodes, and D_x is the target-state dimension. The future prediction target is

$$\mathbf{Y}_{t+1:t+H} \in \mathbb{R}^{H\times N\times D_y}, \tag{2}$$

where H is the forecast horizon and D_y is the output dimension.

We distinguish two types of exogenous variables. Historical ExG is available over the look-back window:

$$\mathbf{E}^h_{t-T+1:t} \in \mathbb{R}^{T\times D_h}, \tag{3}$$

and describes external or contextual conditions observed together with the target history. Future ExG is available over the forecast horizon:

$$\mathbf{E}^f_{t+1:t+H} \in \mathbb{R}^{H\times D_f}, \tag{4}$$

and represents known or forecastable future drivers. This separation is important because historical ExG estimates the current latent state, whereas future ExG conditions the unknown horizon. In spatio-temporal node prediction, historical ExG may include external variables and time features, while future ExG may include exogenous forecasts.

The formulation assumes an operational forecasting setting. At time t, the model has access to historical target states and historical ExG up to t, and it has access to future ExG for $t + 1$ to $t + H$ only if those variables are known, scheduled, or forecast by an external source. The future target values are never used as inputs. This assumption is important for preventing leakage: variables that are measured only after the target horizon should not be included in $\mathbf{E}^f_{t+1:t+H}$. If an application lacks reliable future ExG, the same formulation can be used by setting $\mathbf{E}^f$ to calendar features, scenario forecasts, or a learned uncertainty representation, but the resulting claim should be stated as a different forecasting setting.

The graph relation is also allowed to be partially observed. In some systems, A is a physical adjacency matrix; in others, it is a distance kernel, a correlation prior, or an empty prior that is refined by adaptive learning. We therefore separate A from M_G. The matrix A specifies direct relation priors used by graph propagation, while M_G collects auxiliary metadata such as node coordinates, node attributes, or environmental descriptors. This separation lets the same method support both fixed-topology and adaptive-topology forecasting.

To simplify notation, we encapsulate the set of input drivers available at time t into a single tuple $\mathcal{X}_t$:

$$\mathcal{X}_t \triangleq \big(\mathbf{X}_{t-T+1:t}, \mathbf{E}^h_{t-T+1:t}, \mathbf{E}^f_{t+1:t+H}, A, M_G\big). \tag{5}$$

The forecasting model is defined as a parameterized function f_θ:

$$\hat{\mathbf{Y}}_{t+1:t+H} = f_\theta(\mathcal{X}_t), \tag{6}$$

where M_G denotes graph metadata such as coordinates, node attributes, or physical distances. The model should satisfy three requirements. First, it should extract temporal representations from non-stationary node histories. Second, it should propagate information across graph nodes without assuming that all spatial dependencies share one mechanism. Third, it should use future ExG only through variables available at inference time.

When domain constraints are available, the model can be decomposed into an unconstrained spatio-temporal backbone $\mathcal{H}_\theta$ and a constraint operator $\mathcal{C}_\Omega$:

$$\hat{\mathbf{Y}}_{t+1:t+H} = \mathcal{C}_\Omega\big(\mathcal{H}_\theta(\mathcal{X}_t)\big), \tag{7}$$

where Ω denotes domain knowledge. In spatio-temporal node prediction, Ω includes non-negativity, capacity limits, and regime feasibility. In other domains, Ω may encode speed limits, conservation constraints, bounded sensor ranges, or operational safety limits.

Given a training set $\mathcal{D}$, the supervised objective is

$$\min_\theta \frac{1}{|\mathcal{D}|} \sum_{(\mathcal{X},\mathbf{Y})\in\mathcal{D}} \ell\big(f_\theta(\mathcal{X}), \mathbf{Y}\big) + \lambda_\Omega \mathcal{R}_\Omega\big(f_\theta(\mathcal{X}), \mathbf{Y}\big), \tag{8}$$

where $\ell(\cdot)$ is a forecasting loss, $\mathcal{R}_\Omega$ is an optional constraint-aware regularizer, and λ_Ω controls its strength. The formulation does not require a specific graph operator; any module that maps temporal node states to graph-enhanced node states can be used as the graph adapter.

IV. Method

PhyST-Net is an ExG-aware spatio-temporal graph Transformer designed for network-structured node forecasting with known external drivers and domain constraints. Given the inputs defined in Section III, the model follows the pipeline in Fig. 3. Historical target states and historical ExG are first embedded into node-level temporal representations. AGSP converts the temporal history into continuous tokens. A Transformer backbone learns temporal dependencies. R-STMF serves as a graph adapter that maps temporal node states

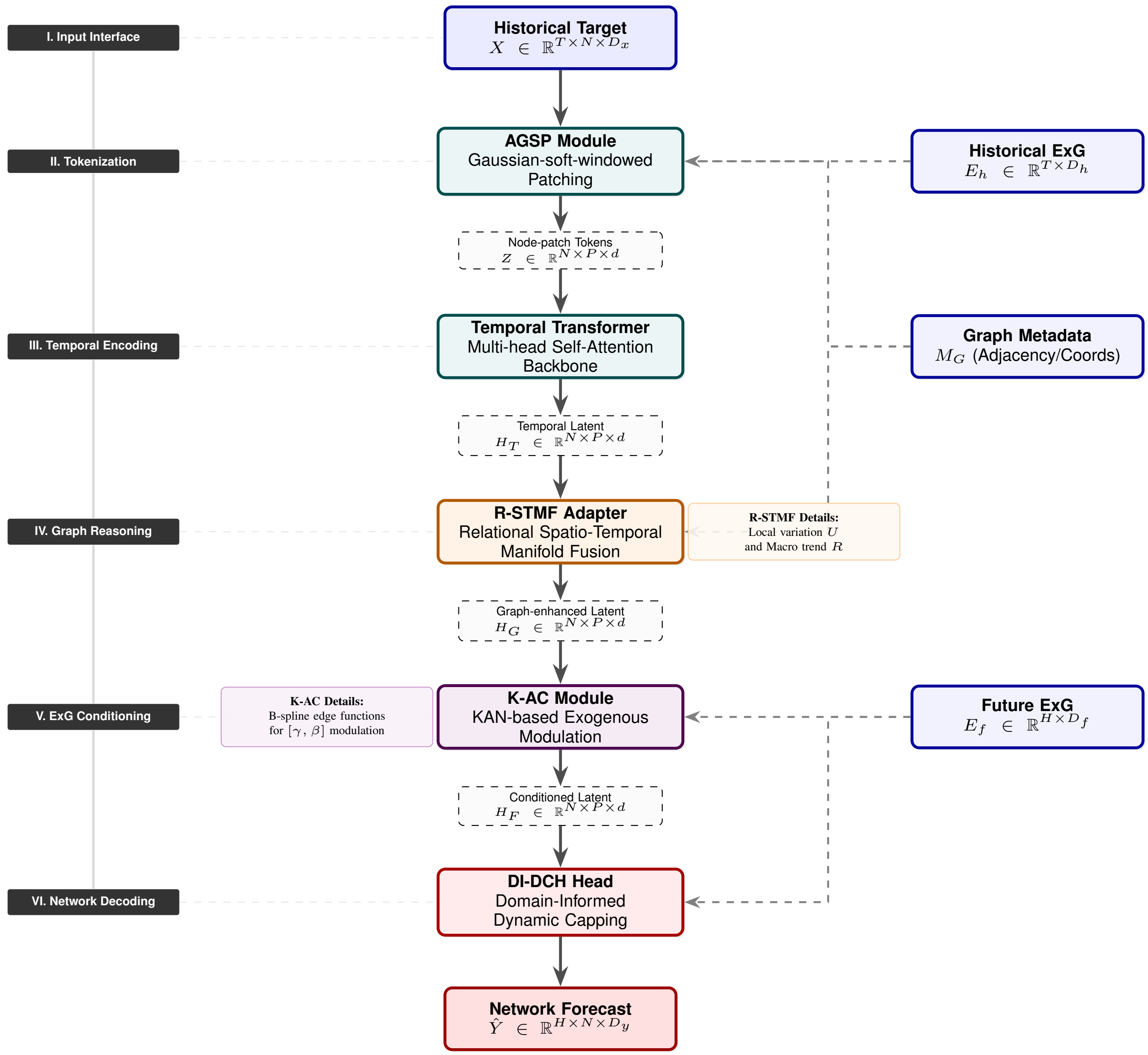

Fig. 3. Overall architecture of PhyST-Net. The framework operates as an exogenous-aware vertical pipeline: (I) Multi-modal input interface; (II) AGSP continuous tokenization; (III) Transformer sequence encoding; (IV) R-STMF relational manifold fusion; (V) K-AC spline-based conditioning; and (VI) DI-DCH domain-constrained forecasting. All external data (ExG and Metadata) and technical details are organized in peripheral tracks to ensure a clear and interpretable data flow.

to graph-enhanced node states. K-AMC injects future ExG into the latent representation. PI-DCH finally decodes the conditioned states into feasible forecasts respecting domain-specific bounds.

A. Overall Framework

For a mini-batch, let $\mathbf{X} \in \mathbb{R}^{B\times T\times N\times D_x}$ denote historical target states, $\mathbf{E}^h \in \mathbb{R}^{B\times T\times D_h}$ denote historical ExG, and $\mathbf{E}^f \in \mathbb{R}^{B\times H\times D_f}$ denote future ExG. The model produces $\hat{\mathbf{Y}} \in \mathbb{R}^{B\times H\times N\times D_y}$. The computation can be written as

$$Z = \Phi_{patch}(\mathbf{X}, \mathbf{E}^h, M_G), \tag{9}$$

$$H_T = \Phi_{temp}(Z), \quad H_G = \Phi_{graph}(H_T, A, M_G), \tag{10}$$

$$H_F = \Phi_{exg}(H_G, \mathbf{E}^f), \quad \hat{\mathbf{Y}} = \Phi_{head}(H_F, \mathbf{E}^f, \Omega), \tag{11}$$

where Φ_{patch} is AGSP, Φ_{temp} is the temporal Transformer encoder, Φ_{graph} is the graph adapter, Φ_{exg} is future-ExG conditioning (K-AMC), and Φ_{head} is the physics-informed prediction head (PI-DCH). This decomposition emphasizes that the architectural backbone is defined on generic graph-structured sequences, while domain-specific knowledge is encapsulated in the input interface and the constraint head.

B. Adaptive Gaussian-soft-windowed Patching

Transformer-based forecasters conventionally mitigate sequence length by aggregating observations into discrete temporal patches. While computationally convenient, fixed non-overlapping patches inevitably induce boundary artifacts when the underlying physical signal transverses a patch boundary. Aligning with emerging principles in complexity-balanced

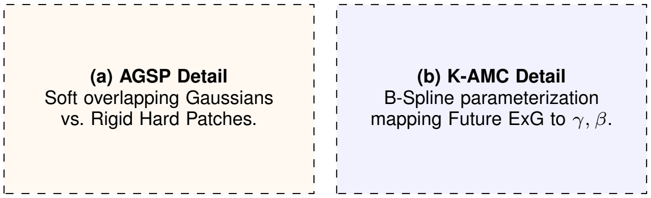

Fig. 4. Internal mechanics of highly coupled modules. (a) AGSP replaces discrete sequence chunking with continuous, differentiable density windows, mitigating boundary artifacts. (b) K-AMC leverages Kolmogorov-Arnold Networks (KAN) to map future exogenous features into dynamic scale (γ) and shift (β) parameters for conditional modulation.

sequence modeling—where representational capacity is dynamically distributed proportional to signal variation—AGSP supersedes rigid patch assignments with learnable Gaussian soft-windows. This differentiable formulation empowers the architecture to autonomously allocate dense receptive fields to highly transient events and broader, smoother fields to stable operational regimes.

For node n and temporal window i, the unnormalized weight at time index $\tau \in \{1, \ldots, T\}$ is

$$W_i(\tau) = \exp\left(-\frac{(\tau - \mu_i)^2}{2\sigma_i^2 + \epsilon}\right), \tag{12}$$

where μ_i and σ_i are learnable center and bandwidth parameters. To ensure numerical stability and valid window properties, we parameterize the bandwidth as $\sigma_i = \text{softplus}(\rho_i) + \delta$, where ρ_i is the learnable parameter. The centers μ_i are initialized to uniform grid locations across the historical window to encourage an ordered temporal partition. The weights are normalized across P windows:

$$\tilde{W}_i(\tau) = \frac{W_i(\tau)}{\sum_{j=1}^{P} W_j(\tau) + \epsilon}. \tag{13}$$

Let $\bar{x}_n(\tau)$ denote the embedded historical state. The token for node n and patch i is

$$z_{n,i} = \psi\left(\sum_{\tau=1}^{T} \tilde{W}_i(\tau)\bar{x}_n(\tau)\right) + e_n + p_i, \tag{14}$$

where $\psi(\cdot)$ is a linear projection, e_n is a node embedding, and p_i is a temporal patch embedding. This formulation allows the model to optimize the temporal receptive fields directly from the forecast loss.

C. Temporal Representation Learning

The token tensor Z is passed to a Transformer encoder to learn dependencies among the P tokens of each node:

$$H_T = \text{STEnc}(Z), \quad H_T \in \mathbb{R}^{B \times N \times P \times d}, \tag{15}$$

D. Relational Spatio-Temporal Manifold Fusion

R-STMF serves as the graph adapter, mapping H_T to H_G. In many systems, spatial dependence contains multiple mechanisms. R-STMF addresses this by decomposing the temporal state along the patch axis:

$$U = \text{DWConv}_\tau(H_T), \quad R = H_T - U, \tag{16}$$

where DWConv_τ is a depth-wise temporal convolution. In our framework, U and R are treated as components capturing local-variation-like and lower-frequency residual features, respectively. For each branch $c \in \{U, R\}$ and patch p, relational attention scores are computed as

$$S_c^{(p)} = \frac{Q_c^{(p)}(K_c^{(p)})^\top}{\sqrt{d_m}} + \mathcal{B}_0, \tag{17}$$

where $\mathcal{B}_0$ is a graph prior bias.

1) Relational Pruning and Masking: To enforce spatial locality and reduce computation, R-STMF employs a Top-K pruning strategy. The attention matrix $A_c^{(p)}$ is computed by masking out elements that are not among the K_c largest scores:

$$\text{Mask}(S, K)_{ij} = \begin{cases} S_{ij} & \text{if } S_{ij} \in \text{TopK}(S_i, K) \\ -\infty & \text{otherwise} \end{cases}, \tag{18}$$

$$A_c^{(p)} = \text{Softmax}\left(\text{Mask}(S_c^{(p)}, K_c)\right). \tag{19}$$

This ensures that node i only aggregates information from its K most relevant neighbors, acting as a structural regularizer against transient noise.

E. Future-ExG Conditioning (K-AMC)

K-AMC injects future exogenous drivers $\mathbf{E}^f \in \mathbb{R}^{B \times H \times D_f}$ into the latent representations. A critical step is aligning the horizon-wise drivers with the patch-wise latent states. We define an alignment operator $\Gamma : \mathbb{R}^{H \times D_f} \to \mathbb{R}^{P \times d_{exg}}$ that maps the H future steps into P exogenous tokens via temporal pooling or a linear projection. The aligned representation $\bar{\mathbf{E}}^f$ is then passed through a KAN block to produce modulation parameters:

$$[\gamma, \beta] = \text{KAN}(\bar{\mathbf{E}}^f), \quad \gamma, \beta \in \mathbb{R}^{B \times N \times P \times d}. \tag{20}$$

The conditioned latent state is then obtained via element-wise scale and shift:

$$H_F = H_G + \lambda\left(\gamma \odot H_G + \beta - H_G\right). \tag{21}$$

F. Physics-Informed Dynamic Capping Head (PI-DCH)

PI-DCH instantiates the constraint operator $\mathcal{C}_\Omega$. In mission-critical environments such as smart grids, safety-critical traffic routing, and industrial control systems, deploying generative or predictive models mandates rigorous output-space guidance to preclude physically unfeasible trajectories. PI-DCH achieves this by systematically steering latent predictions toward a dynamic feasible envelope C_{dyn} prior to the enforcement of a strict physical boundary.

Specifically, a linear decoder first generates an unconstrained predictive tensor $\tilde{Y} = \text{Linear}(H_F)$. Concurrently, a learned barrier gate $u = \sigma(g(H_F))$ modulates this prediction smoothly:

$$\hat{Y}_{soft} = \tilde{Y} + u \odot (C_{dyn} - \tilde{Y}). \tag{22}$$

During the training phase, this soft-capped tensor $\hat{Y}_{soft}$ is utilized to compute the forecasting loss, guaranteeing that gradients backpropagate natively through the constraint-shaping logic. The definitive feasible forecast $\hat{Y}$ is subsequently derived via a non-linear projection:

$$\hat{Y} = \text{clip}(\hat{Y}_{soft}, 0, C_{dyn}). \tag{23}$$

This physics-informed projection is stringently enforced during inference, mathematically guaranteeing absolute operational feasibility without resorting to opaque, post-hoc heuristic corrections.

G. Computational Complexity Analysis

A core motivation of PhyST-Net is mitigating the severe computational bottlenecks of dense spatio-temporal Transformers, which conventionally exhibit a time complexity of $\mathcal{O}(N \cdot T^2 + T \cdot N^2)$ for sequence length T and node count N. PhyST-Net structurally resolves this through its decoupled topology.

In the temporal pathway, the AGSP module compresses the raw T-length sequence into P continuous tokens (where $P \ll T$). Consequently, the temporal Transformer encoder's self-attention complexity is reduced from $\mathcal{O}(N \cdot T^2)$ to $\mathcal{O}(N \cdot P^2)$.

In the spatial pathway, standard full-graph attention requires $\mathcal{O}(P \cdot N^2)$ operations. The R-STMF module introduces a Top-K sparse masking strategy (Eq. 19), restricting message passing to a parameterized neighborhood size K. This bounds the spatial complexity to $\mathcal{O}(P \cdot K \cdot N)$. Thus, the overall self-attention complexity of the PhyST-Net backbone is decoupled and bounded by $\mathcal{O}(N \cdot P^2 + P \cdot K \cdot N)$, rendering the architecture highly scalable for large physical networks and extended forecasting horizons without sacrificing expressive capability.

H. Optimization Objective

To bridge the gap between purely data-driven forecasting and physical feasibility, we minimize a multi-objective loss $\mathcal{L} = \mathcal{L}_{mse} + \lambda_{\Omega}\mathcal{L}_{penalty}$. The penalty term $\mathcal{L}_{penalty}$ structurally regularizes the model to internalize the operational set Ω during backpropagation. Using the positive-part operator $[x]^+ \triangleq \max(0, x)$, the penalty is defined as:

$$\mathcal{L}_{penalty} = \frac{1}{NH}\sum_{n,t}\Bigg(\underbrace{[-\hat{Y}_{n,t}]^+}_{\text{Non-negativity}} + \underbrace{[\hat{Y}_{n,t} - C_{dyn,n,t}]^+}_{\text{Capacity Limit}}\Bigg). \tag{24}$$

This term ensures the model learns to respect physical boundaries natively, thereby reducing the reliance on reactive clipping at inference time. The overall training logic is summarized in Algorithm 1.

Algorithm 1 PhyST-Net Forward Pass

Require: $\mathbf{X}, \mathbf{E}^h, \mathbf{E}^f, A, M_G$.
Ensure: $\hat{\mathbf{Y}}$.
Embed historical target states and historical ExG.
Construct AGSP tokens Z using Eqs. (13)–(14).
Encode temporal dependencies to obtain H_T.
Apply R-STMF graph adaptation to obtain H_G.
Inject future ExG through K-AMC to obtain H_F.
Decode with PI-DCH using Eqs. (22)–(23).
return $\hat{\mathbf{Y}}$.

V. Experiments

VI. Discussion

VII. Case Studies

VIII. Conclusion

[Placeholder: Write the final conclusion after the experimental section is complete.]

The final conclusion should summarize the problem setting, the ExG-aware STGT formulation, the roles of AGSP, R-STMF, K-AC, and DI-DCH, and the main validated findings from SDWPF, HoT, and Kelmarsh. It should also state the remaining limitations, such as cross-domain transfer, operational uncertainty, probabilistic forecasting, and larger-scale deployment.

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